

Video And Graphics

3D Pipeline

The process of 3D graphics can be divided into three-stages: tessellation, geometry and rendering. In the tessellation stage, a described model of an object is created, and the object is converted to a set of polygons. The geometry stage includes transformation, lighting, and setup. The rendering stage, which is critical for 3D image quality, creates a two dimensional display from the polygons created in the geometry stage.

Alpha Buffer

An extra colour channel to hold transparency information. In a 32-bit frame buffer there are 24 bits of colour, 8 each for red, green, and blue, along with an 8-bit alpha channel.

Anti-aliasing

Anti-aliasing is sub-pixel interpolation—a technique that makes edges appear to have better resolution.

Atmospheric Effect

Effects, such as fog and depth cueing, which improve the rendering of real-world environments.

Bitmap

A pixel-by-pixel image.

Blending

The combining of two or more objects by adding them on a pixel-by-pixel basis.

Double Buffering

A method of using two buffers, one for display and the other for rendering. While one of the buffers is being displayed, the other buffer is operated on by a rendering engine. When the new frame is rendered, the two buffers are switched. The viewer sees a perfect image all the time.